Development of SnO_2@C nanocomposites derived from cassava residues as a green anode for advanced lithium-ion batteries

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Highlights

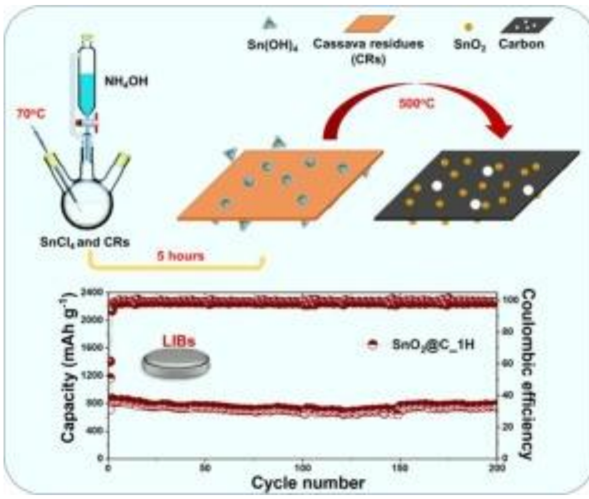
- SnO_2@C nanocomposites from cassava residues were prepared by a simple method.
- The SnO_2 average particle sizes in the SnO_2@C nanocomposites ranged from 35 to 55 nm.
- The SnO_2@C_1H electrode showed a capacity of 777.7 mAh g^{-1} at 0.2 A g^{-1} after 200 cycles.
- The SnO_2@C_1H electrode achieved a capacity of 738.1 mAh g^{-1} at 5 A g^{-1}.
- The coulombic efficiency of SnO_2@C electrodes approached 100% from the 2nd cycle.

Abstract

In this report, nanocomposites of SnO_2 and carbon (SnO_2@C nanocomposites) were synthesized by a solution-based method combined with calcination at 500 °C in Ar for 1 or 2 h, utilizing $SnCl_4 \cdot 5H_2O$ and cassava residues as the Sn and C precursors, respectively. Due to this unique structure, high first-cycle charge capacities of 716.6 and 837.2 mAh g^{-1} at 0.2 A g^{-1} were achieved for SnO_2@C electrodes calcined for 1 h and 2 h, namely SnO_2@C_1H and SnO_2@C_2H, respectively. Furthermore, the SnO_2@C_1H electrode maintained a capacity of 777.7 mAh g^{-1} after 200 cycles, with the coulombic efficiency approaching 100 % after the initial cycle. The rate capacity test also demonstrated that the

high current density charging capability of the SnO_2@C electrodes was superior, showing capacities of 738.1 mAh g^{-1} for SnO_2@C_1H and 631.2 mAh g^{-1} for SnO_2@C_2H at 5 A g^{-1}. Last but not least, the pseudo behavior was also confirmed to be consistently higher at all scan rates, elucidating the fast-charging capability of the SnO_2@C electrodes. With the improved performance, the SnO_2@C_1H nanocomposite has demonstrated excellent electrochemical ability and is suitable for anode material of advanced lithium-ion batteries.

Graphical abstract



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Introduction

As energy sources from fossil fuels have gradually been depleted, the need for robust energy storage systems has become indispensable, considering the rapid development of science and technology [1]. In that context, Li-ion batteries (LIBs) have emerged as a strong solution and have been widely used in many electronic devices for decades [2,3]. However, this type of battery is still facing significant challenges, including the requirement for greater energy storage capacity to improve performance when used in electric vehicles and large-scale devices [4,5]. One of the reasons for the limited capacity of commercial LIBs today lies in the graphite anode material, which has a low theoretical capacity (~372 mAh g^{-1}) [6]. Hence, several studies have been carried out to improve the capacity of anode materials. Promising candidates based on metal-derived active materials, such as Sn, Bi, and Sb, have been discovered. These metals can react with Li^+ ions via both the conversion and alloying mechanisms, thereby increasing the number of charges during the reactions and, consequently, increasing the capacity of the electrodes [[7], [8], [9]]. However, metal-based electrode materials suffer from significant volume expansion during operation [10], which leads to internal stress and structural degradation of the electrode during battery cycling [11,12]. To overcome this drawback, some studies have proposed using carbon-based materials combined with metal oxides to minimize the volume expansion of metal-based electrodes [13]. In these composites, the carbon acts as a buffer agent that reduces structural stress under significant volume changes of the metal component during battery operation [14,15]. For example, the study of Zhang et al. [16] on iron oxide-based electrode materials showed that the electrode material with carbon fibers had better cycling stability with a specific capacity after 100 cycles of 709.0 mAh g^{-1}, while the pure metal oxide material had a specific capacity of only 187.2 mAh g^{-1}. In another study, Xie and colleagues [17] reported that using carbon-based materials to overcome the volume expansion of metal-based electrode materials is feasible, thereby improving the stability of the electrode materials. The results showed that after 100 cycles, the composite material from rGO and metal oxide had a capacity of 498.2 mAh g^{-1}, much higher than only 78.5 mAh g^{-1} of pure metal oxide materials.

Among the metal-based electrode materials, tin dioxide has attracted much attention because of its outstanding theoretical capacity, about 1493 mAh g^{-1} [18]. Unfortunately, like metal-based electrode materials, SnO_2 has a 3-fold volume expansion when reacting with Li^+ ions. For instance, SnO_2 nanosheet developed by D. Narsimulu et al. [19] demonstrated high capacity, reaching 1350 mAh g^{-1} in the first discharge cycle; however, the capacity of this electrode rapidly degraded to 258 mAh g^{-1} at 0.1 A g^{-1} after 50 cycles. Recent studies have shown promising results on the composite ability of carbon materials with SnO_2, aiming to minimize the volume expansion of SnO_2-based electrode materials. For example, the combination of SnO_2 and carbon nanotubes [18] improved the cycling capacity of LIBs, exhibiting a capacity of 809 mAh g^{-1} at 0.2 A g^{-1} after 100 cycles. Although they have made significant contributions to the improvement of the cycling durability of SnO_2-based anodes, carbon materials such as CNTs [20], graphene [21], and N-doped carbon [22] generally require relatively complicated fabrication methods, which makes it challenging to apply these anodes in commercial production. Given this situation, the application of carbon from biomass as an electrode material is being widely studied [23]. Carbon from biomass sources is also considered a potential candidate to help improve the performance of LIBs [24]. Moreover, it offers several advantages, such as low cost, non-toxicity, and abundant raw materials. In particular, utilizing biomass derived from agricultural and industrial residues helps mitigate environmental pollution by reducing waste discharge [25,26]. Cassava residue represents one of the most abundant sources of biomass, as cassava is a major food crop widely cultivated for human consumption [27]. Industrial processing of cassava, such as starch production, animal feed, and pharmaceuticals, generates substantial quantities of waste [28,29]. Approximately 350 million metric tons of cassava-derived waste, including peels, stems, leaves, roots, and processing residues, are discharged annually [27]. Recent studies highlight cassava waste as an ideal raw material for carbon material synthesis. For example, Jordana Georgin and colleagues [30] successfully synthesized activated carbon derived from cassava peels, which is utilized for wastewater treatment and exhibits an herbicidal absorption efficiency of approximately 68 %. In a separate study, Sulaiman and colleagues [31] used cassava stems as a precursor to synthesize adsorbent materials for removing Ofloxacin from pharmaceutical waste. Beyond waste treatment, cassava waste is also explored for applications in devices such as supercapacitors and lithium-ion batteries [32]. Using cassava residue as a precursor, Chau and colleagues [33] successfully synthesized electrode materials for lithium-ion batteries.

Therefore, in this study, we propose the use of cassava residues (CRs) – a by-product of cassava starch production in the food industry – as a biomass source for carbon material synthesis. The combination of SnO_2 and carbon derived from CRs was prepared by a solution-based method, followed by calcination at 500 °C for 1 or 2 h under Ar gas flow and named SnO_2@C_1H and SnO_2@C_2H, respectively. The SnO_2@C_1H anode electrode demonstrated good cycling ability and high capacity by maintaining a capacity of 777.7 mAh g^{-1} after 200 cycles at 0.2 A g^{-1} for LIBs. The use of CRs as a raw material not only helps address environmental waste and reduce production costs, but also enables the development of high-performance electrode materials with commercial potential as alternative anodes for LIBs in the future.

Materials

Cassava residues (CRs), collected from a cassava starch factory in Gia Lai province, Vietnam, were used as the raw material for nanocomposite synthesis. The CRs were crushed and dried at 90 °C for 12 h prior to use. Stannic chloride pentahydrate ($SnCl_4 \cdot 5H_2O$, 99.0 %) and ammonia solution (NH_4OH, 25 %) were purchased from Pallav Chemicals & Solvents Pvt. Ltd (Mumbai, India).

Synthesis of SnO_2@C nanocomposites

SnO_2@C nanocomposites were prepared from $SnCl_4 \cdot 5H_2O$ and CRs precursors. First, a 100 ml solution of 0.6 M $SnCl_4$ was

Results and discussions

The formation of SnO_2@C nanocomposites (Fig. 1) can be described as follows: First, the precursors reacted to form $Sn(OH)_4$ precipitate (Eq. (1)), then the precipitated species adsorbed onto the surface of CRs. Next, heat treatment converted the CRs into carbon (Eq.

(2)) and pyrolyzed $Sn(OH)_4$ to form SnO_2 (Eq. (3)).$SnCl_4 + 4\ NH_4OH \rightarrow Sn(OH)_4 + 4\ NH_4Cl$CRs → CO + C + $H_2O$$Sn(OH)_4 \rightarrow SnO_2 + 2\ H_2O$

Fig. 2 presents the XRD patterns of the SnO_2@C_1H and SnO_2@C_2H nanocomposites and the standard patterns of

Conclusion

In summary, SnO_2@C nanocomposites were successfully prepared by a simple method, employing $SnCl_4$, NH_4OH, and cassava residues as the only reagents and precursors. The morphology, particle size, and phase composition of SnO_2@C nanocomposites were adjusted by varying the calcination time from 1 to 2 h at 500 °C in Ar gas. XRD, Raman, and EDS methods demonstrated the existence of SnO_2 and carbon phases in the synthesized materials. In addition, the morphology of SnO_2@C materials observed through

CRediT authorship contribution statement

Bui Thi Thao Nguyen: Methodology, Investigation. **To Giang Tran:** Resources, Formal analysis. **Phan Vien Nguyen:** Resources, Methodology. **Nguyen Huu Huy Phuc:** Visualization, Validation. **Pham Trung Kien:** Methodology, Formal analysis. **Nhi Tru Nguyen:** Writing – original draft. **Liem Thanh Pham:** Resources, Methodology. **Man Van Tran:** Resources, Formal analysis. **Tuan Loi Nguyen:** Writing – review & editing, Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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